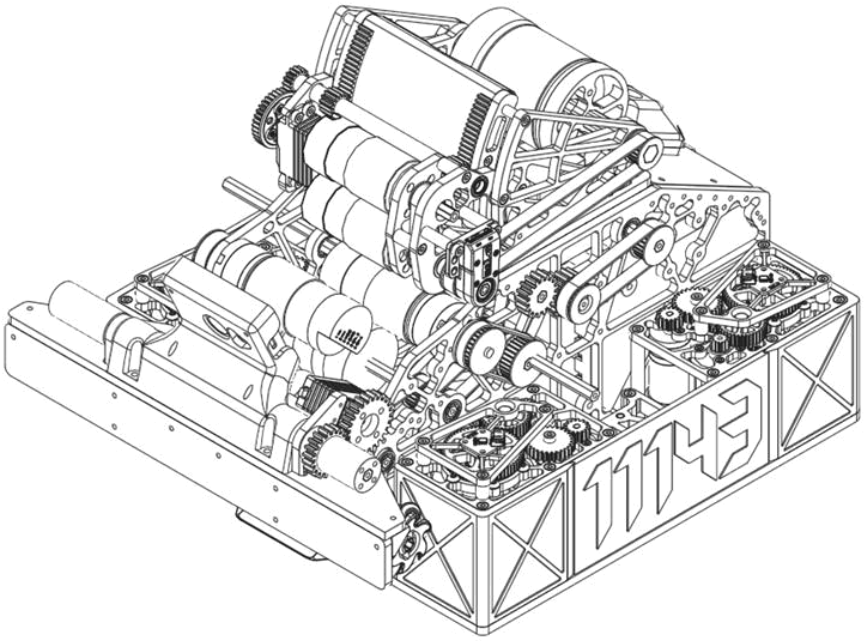
ENGINEERING PORTFOLIO

FTC - FIRST TECH CHALLENGE

BARKER YELLOWBACKS

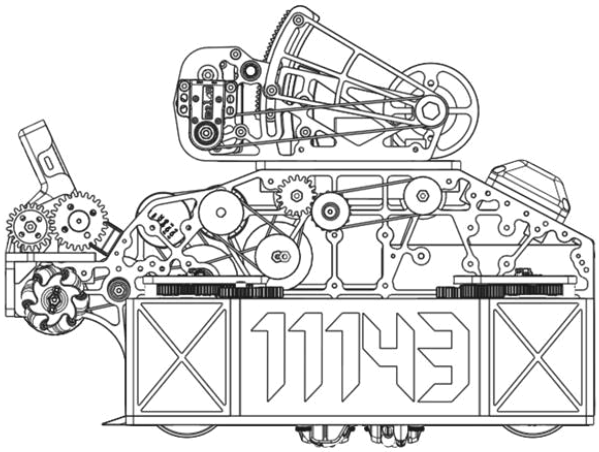
11143



2026-2027

Sydney
Australia

Barker College
Supported by Barker Robotics



MEET THE TEAM

13 MEMBERS

EB — **Ethan B.**
LEAD DESIGN
assembly

BL — **Brandon L.**
CAD
assembly

IH — **Isaac H.**
LEAD CODE
CAD, assembly

RT — **Ryan T.**
DRIVER
CAD

WQ — **Will Q.**
LEAD STRATEGY
drive coach

IW — **Ivy W.**
PROGRAMMING
HR

TX — **Tina X.**
CODE
assembly

CW — **Chelsea W.**
CODE
assembly

CL — **Chloe L.**
ELECTRICAL
assembly

CS — **Chloe S.**
STRATEGY
code, assembly

CC — **Cooper C.**
SCOUTING
assembly

JC — **Jemima C.**
PROGRAMMING
scouting

LT — **Lucas T.**
CAD
assembly

HOW WE SPLIT THE WORK

- **CAD:** 4 members in Onshape, with our lead designer
- **Code:** 6 members write and test the code
- **Strategy:** 2 strategists plus 2 scouts
- **Assembly:** 9 of 13 members build

HOW WE WORK

- We meet **every week or two**
- Each specialty decides its own area (mechanics decide mechanical changes), then tells the whole team
- Each subsystem has its own code branch, merged into one robot

OUR SEASON

2026 SO FAR

23 SEP	12 OCT	20 OCT	25 OCT	12 NOV
Shooter	**No turret**	**Intake**	**One robot**	**Wollongong**
first shooter code	turret removed	intake and transfer	subsystems merged	first event

WHAT WE KEPT FROM LAST SEASON

Last season's robot (background only) taught us:

- **Swerve works for us:** we kept it
- **A turret takes months to tune:** April to June 2026, so this year we dropped it
- Logging and replay carried over to this season's code

THIS SEASON'S GOALS

- Shoot from **anywhere** on the field
- **Less waiting** between shots
- Collect along the wall and shoot **without turning round**
- Hold our spot against **defence**
- Keep the robot **light**, so it moves more efficiently

WHERE WE ARE: ROBOT NOT BUILT YET

DATE	WHAT WE FINISHED (FROM OUR CODE HISTORY)
22-23 Sep	Code repository and simulator set up; a folder for each subsystem
6 Oct	Speed controller (PIDF) working for the shooter
7 Oct	Hardware and controller bindings added
12 Oct	Turret removed; hood angle limits added; lookup table started
16 Oct	Lookup table can work out values between its points
20-23 Oct	Intake and transfer code finished
25 Oct	Shooter and intake merged; robot CAD in the simulator; 3 of 3 scored in simulation
26 Oct	Simulated pollen now rolls, bounces and makes contact with the robot

OUTREACH

7	Fortnightly	1,500+	About 32
members teach Darkinjung students	since about February 2026	people at Round Square	countries at Round Square

TEACHING DARKINJUNG STUDENTS

WHAT

7 members of our team teach robotics to Darkinjung students **once a fortnight**, since about February 2026.

WHAT THEY LEARN

They built robots for this year's **VEX IQ** game and learnt how to build, how to code, how to do interviews and how to write portfolios.

IMPACT

Some of the students were in competitive teams and won **Skills**, **Teamwork Champion** and **Excellence** awards.

ROUND SQUARE INTERNATIONAL CONFERENCE

WHAT

On **28 September 2026** about **7** of us ran the robotics part of the conference at Barker.

REACH

We showed our robotics program to **over 1,500 people** from around **32 countries**, and introduced them to FIRST.

MENTORING VEX TEAMS

Ethan B. and Isaac H. mentor VEX teams.

BARKER VEX COMPETITION

Ethan B., Isaac H. and Brandon L. volunteered.

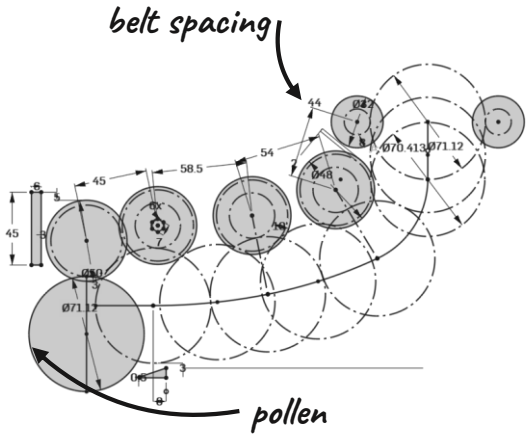
PHOTOS

<photos of the Darkinjung sessions and Round Square>

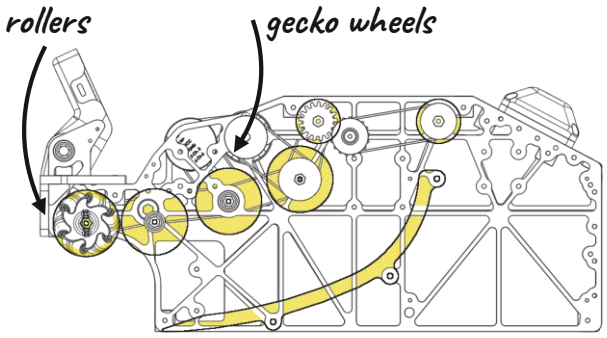
OUR DESIGN LOOP

1	2	3	4	5
Goals	**Sketch + CAD**	**Simulate**	**Build**	**Test + change**
what the game needs	Onshape layouts	same code, desktop sim	parts and assembly	back to step 2

EXAMPLE: FROM SKETCH TO PART



1. Layout sketch first: pollen size, roller sizes and belt spacing (mm)



2. Then the plates and belts are designed around it in CAD

ONE ROBOT, MANY BRANCHES

Each subsystem was coded on its own branch: **intake** (Tina X.), **shooter** (Jemima C.), then merged into one robot on **25 October**.

What merging taught us

- Two branches each hid the other's subsystem: fixed
- Units were mixed (RPM, ticks, degrees): one unit per value now
- All the fixes are on page 12

IDEAS WE BROUGHT IN

FROM VEX

Ethan B., Will Q., Ryan T. and Tina X. played VEX **Push Back**. The game was very similar, so our matchloader follows the same idea (page 10).

FROM LAST SEASON

Swerve and logging carried over; the turret did not (page 9).

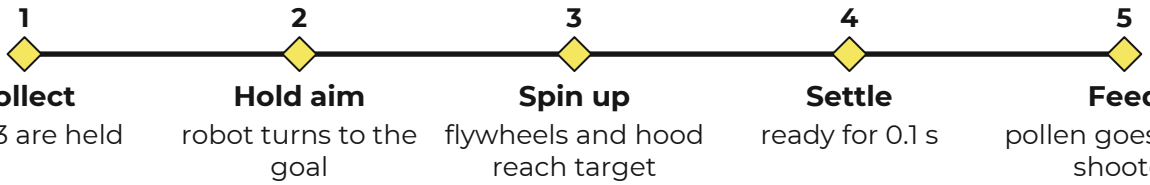
STRATEGY AND REQUIREMENTS

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HOW WE PLAY

<our scoring plan: what we score first, where we shoot from, our role with the alliance partner>
Pollen collects along the wall, so we collect there and shoot from there.

ONE SCORING CYCLE, AS OUR CODE RUNS IT



REQUIREMENTS, AND HOW THE ROBOT MEETS EACH

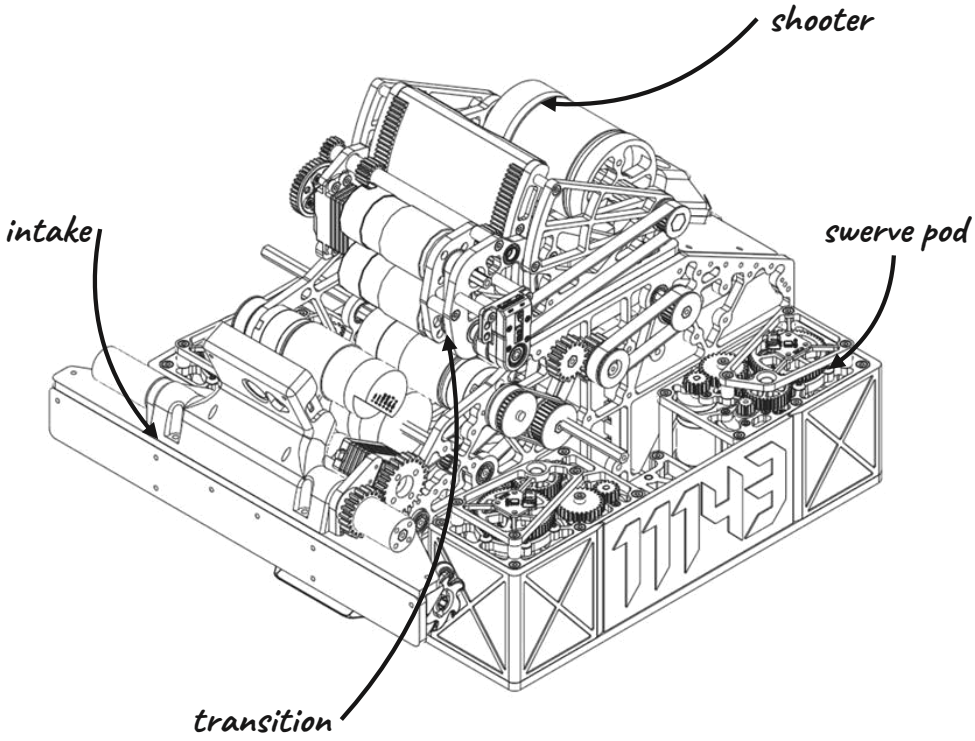
WE NEED TO	WHY	OUR ANSWER
Shoot from anywhere	fewer trips across the field	moving hood
Shoot quickly, again and again	less waiting between shots	two flywheels
Shoot without turning round	pollen collects along the wall	rear-facing shooter
Aim while driving	save time every cycle	swerve + auto-aim
Hold our spot	defence will push us	swerve X-lock
Move efficiently	a big advantage in matches	swerve; a light robot
Load the flower fast	<why>	matchloader
Find pollen and the goal	drive and aim without guessing	front + rear cameras

SCOUTING

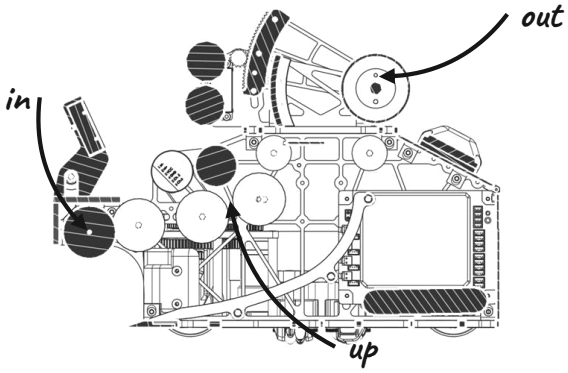
Cooper C. and Jemima C. scout: <what we record and how it changes our plan>

ROBOT OVERVIEW

OUR ROBOT



POLLEN PATH

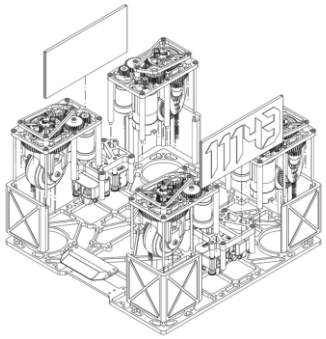


In at the front, up the transition, out of the rear shooter

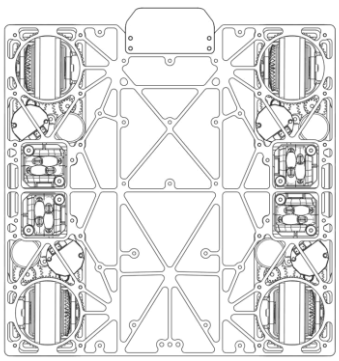
NUMBERS

ITEM	VALUE
Size (CAD)	407.7 x 327.0 x 290.9 mm
Pod centres (CAD)	239 x 239 mm
Wheels (CAD)	60.9 mm
Motors	8: drive 4, intake 2, shooter 2
Servos	8: steering 4, hood, blocker, matchloader, gate
Cameras	2: front, rear

CHASSIS: LIGHT, SO IT MOVES MORE EFFICIENTLY



Exploded: pods, plates and side panels

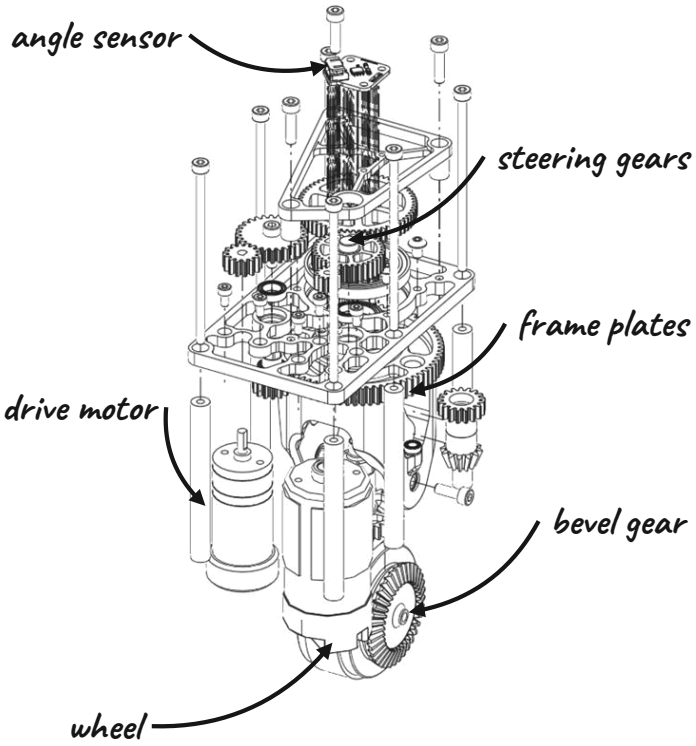


Underneath: pods in the corners

SWERVE DRIVETRAIN

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ONE SWERVE POD



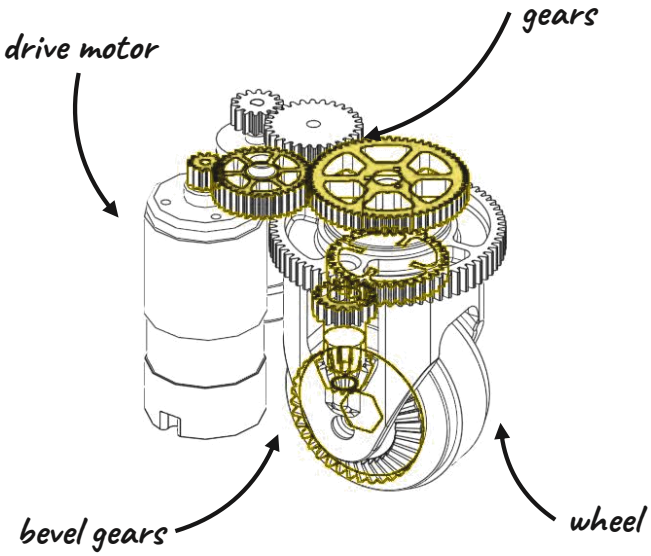
WHY SWERVE, AGAIN

- **X-lock:** the wheels sit in an X, and the friction makes us a lot harder to push or defend
- Moves a lot more **efficiently**: a big advantage in matches
- Full-grip wheels: more **speed and pushing power** than mecanum
- **Aim while driving:** drive one way, face the goal
- We built swerve last season, so the risk was low

Key features

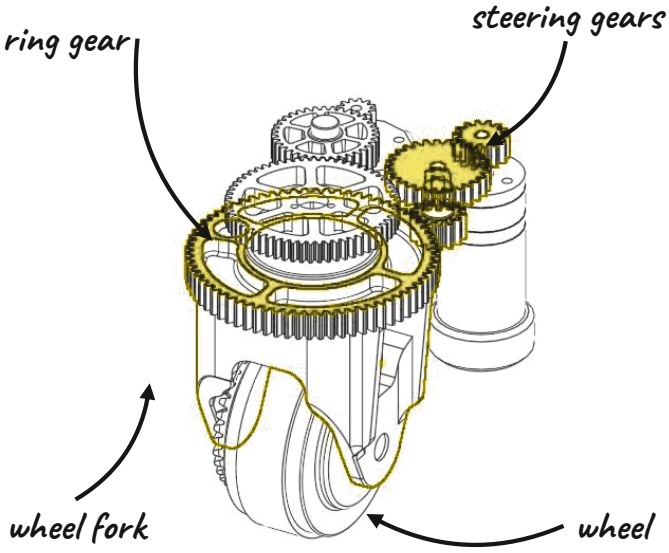
- **4 pods**, each with a drive motor and a steering servo
- An **absolute angle sensor** in every pod, so each wheel knows where it points from the start
- Pod centres **239 x 239 mm** apart; wheels **60.9 mm**

WHEEL DRIVE



Motor, gears, then a bevel gear turn the wheel

MODULE ROTATION



Servo and ring gear turn the whole pod

TESTS

TEST	RESULT	STATUS
Top speed	not built yet	**NOT TESTED**
Pushed by another robot in X-lock	not built yet	**NOT TESTED**

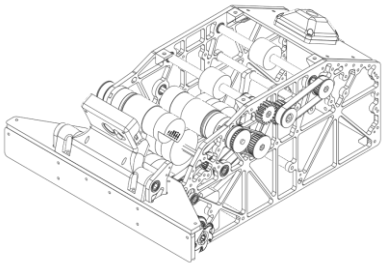
INTAKE AND TRANSITION

WHAT WE TRIED

V1 PRINTED FUNNELS

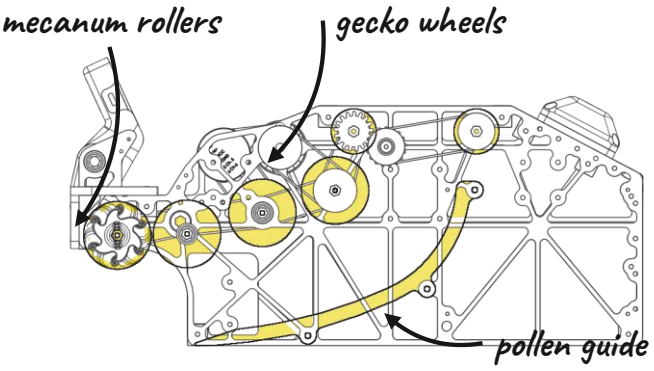
+ <what worked>
− gecko wheels at the front only
− <what went wrong>

V2 ROLLERS + TRANSITION



+ mecanum rollers sweep pollen in
+ gecko wheels grip it and carry it up
− <downside, if any>

POLLEN PATH

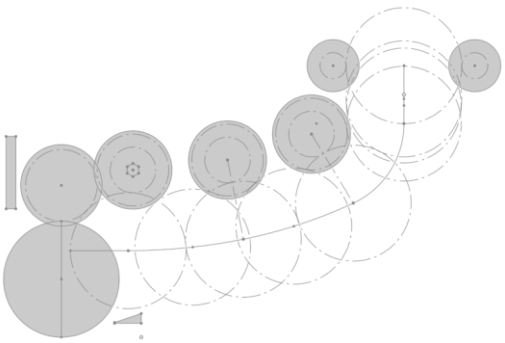


Rollers pull pollen in; gecko wheels lift it to the shooter

KEY SIZES AND SENSING

- Pollen **71.12 mm**; intake rollers **50 mm**
- Gecko wheels **48 mm**
- Small gecko wheels **32 mm** at the top, so less **backspin** goes into the shooter
- Belt spacing **45, 58.5, 54, 44 mm**
- 2 motors drive the intake and transition
- **Polycarbonate ramps** give the pollen a low-friction surface
- **3 beam sensors** count the pollen (empty, one, two, full); the intake stops when full
- Jams: not tested yet (robot not built)

MATHS: WHERE THE POLLEN TOUCHES



Our two rows of gecko wheels sit apart, so each row meets the pollen off its centre. The sketch plane cuts the pollen in a circle of **70.413 mm**:

$d = \sqrt{(35.56^2 - 35.21^2)} = 5.0$ mm

So the plane is 5.0 mm from the pollen's centre, and the sketch uses 70.413 mm there, not 71.12 mm.

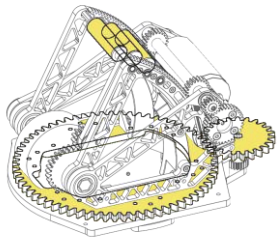
WHAT WE TRIED

TURRET, LAST SEASON

+ aims without turning the robot

– months of tuning (Apr-Jun 2026)

TURRET, THIS SEASON



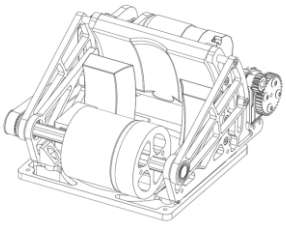
+ what we wanted first

– weight and space

– 1 more motor or servo

– no time to build it

FIXED REAR SHOOTER



+ fewer parts

+ shoots out of the flowers without turning

– the whole robot must turn to aim

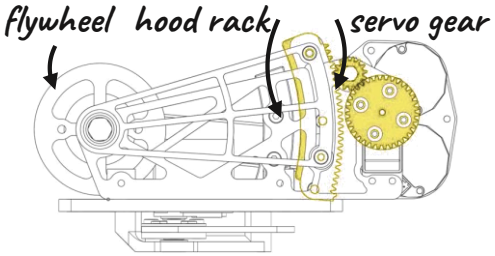
WHAT WE WANTED, AND WHY

- **Two flywheels:** more flywheel means **less waiting** between shots
- **Moving hood:** shoot from anywhere on the field. Range in our code: **33.6 to 67 degrees**
- **Facing backwards:** pollen collects along the wall, so we drive up to the wall and aim, instead of collecting and then turning round
- **Lookup table:** flywheel speed and hood angle from where localisation says we are. The table is started in code; values are measured once the robot is built

What we changed

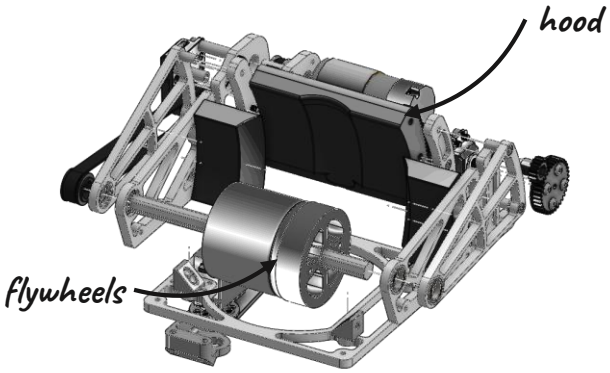
The turret got as far as CAD and code, never built: not enough time. On **12 October** we removed it, and the swerve base turns the robot to aim instead (page 13).

HOOD DRIVE



A servo gear drives a rack to tilt the hood

EXPLODED VIEW



TESTS

TEST	RESULT	STATUS
Shots in from 3 distances	not built yet	NOT TESTED
Time between shots	not built yet	NOT TESTED

MATCHLOADER AND CAMERAS

MATCHLOADER

IDEA

Four of us played VEX **Push Back**, a very similar game. We used the same kind of matchloader for the flower.

PROBLEM

<what the matchloader fixes, with a number>

RESULT

Not tested yet: the robot is not built.

BLOCKER AND GATE

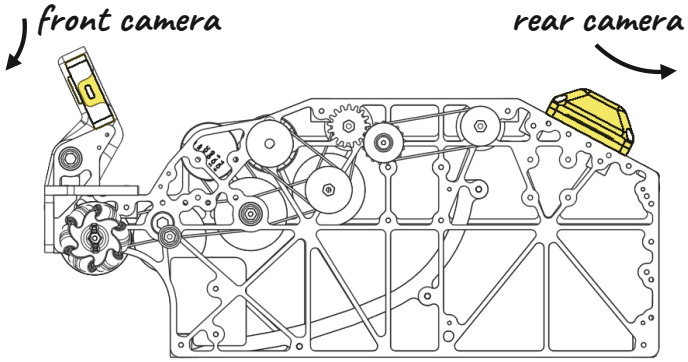
BLOCKER

<what the blocker stops> (1 Axon Max servo)

GATE

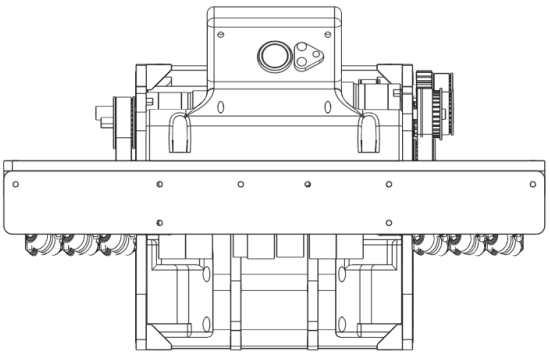
Stops pollen going into the shooter until the shot is ready (1 Axon servo).

TWO CAMERAS



- **Front camera:** finds pollen, so we can path to it
- **Rear camera:** reads AprilTags, so we know where the goal is
- Status: not tested yet (robot not built)

HOW POLLEN IS HELD, THEN RELEASED

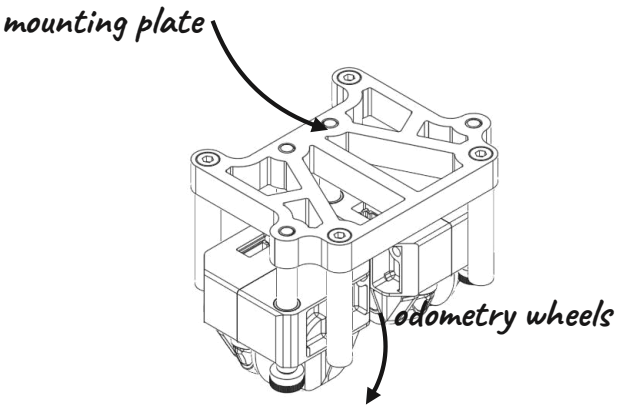


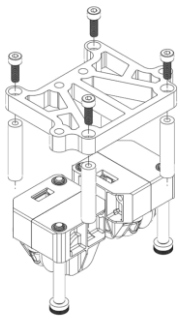
- **Collect:** the intake runs until the 3 beam sensors say we are full, then it stops
- **Hold:** pollen waits while the robot aims
- **Release:** pollen only feeds once aim, flywheel speed and hood have all been ready for **0.1 s** (page 13)

WHAT EACH PART DOES

PART	PROBLEM IT SOLVES	WHAT IT DOES
4 drive motors	moving in any direction	drive each swerve wheel
4 steering servos + 4 angle sensors	pointing every wheel the right way	turn each pod and read its angle
IMU	knowing which way we face	heading for driving and aiming
Front and rear cameras	finding pollen and the goal	pollen pathing; AprilTags
3 beam sensors	knowing how much pollen we hold	sense pollen at bottom, middle and top
Gate servo	pollen reaching the shooter too early	holds pollen until the shot is ready
2 shooter motors	the same shot every time	flywheels held at target speed (PIDF)
Hood servo	shooting from different distances	sets the launch angle
Voltage and current checks	brown-outs and stalled motors	logged every loop

ODOMETRY POD





Designed in CAD; the code uses wheels + IMU for now

SPEED CONTROL

CHALLENGE

Each shot takes energy from the flywheels, and a slower wheel shoots short.

SOLUTION

A **PIDF** controller holds the target speed; feedforward uses battery voltage, so a flat battery does not change the shot.

IN OUR CODE

Flywheels are limited to **3500 RPM** and count as ready within **50 RPM** of the target; the hood within **2 degrees**.

RESULT

Not tested yet: the robot is not built.

SAME CODE ON THE ROBOT AND ON A LAPTOP

PART	WHAT IT HOLDS	WHY
RobotCode	behaviour: driving, aiming, shooting	written once, used everywhere
TeamCode	the real motors, servos and sensors	only place that touches hardware
RobotSim	simulated robot, field and pollen	test before the robot is built

WHY WE SIMULATE

CHALLENGE

The robot is not built yet, but the code needs testing now.

SOLUTION

The same TeleOp runs in our simulator on the official field CAD.

RESULT

Simulation: scores **3 of 3** for both alliances, also while moving.

AUTOMATED TESTS

- **24 desktop tests** pass (25 October 2026)
- They cover heading wrap, 10 ms loops, a lost heading sensor, manual turn override and stop outputs
- Every loop is logged, so a run can be replayed

WHAT WE FOUND AND FIXED WHEN WE MERGED THE CODE

PROBLEM	FIX
Shooter could never reach its shooting state	a target speed now starts the wheel controller
Updates changed numbers but moved nothing	read feedback, write a limited motor power every loop
RPM, encoder ticks, degrees and servo positions mixed	one unit for each value; conversion only in hardware code
The flywheel's controller also drove the hood	the hood has its own mapping and rate limit
Lookup table divided by zero at its first point	handles exact points; never guesses past its ends
Simulated aiming stopped about 12 degrees short	wheel feedforward based on battery voltage

RISKS AND HOW WE REDUCE THEM

RISK	HOW WE REDUCE IT
Simulation is not the real robot	the code refuses to feed until the shooter is calibrated
Heading sensor lost mid-match	tested in simulation: stops driving and feeding
Flat battery changes the shot	feedforward scaled by battery voltage
A complex turret fails	no turret: the drivetrain aims

AUTO-AIM

THE PROBLEM

CHALLENGE

We dropped the turret, so the shooter faces backwards and cannot turn on its own.

SOLUTION

Holding aim turns the **whole robot** until it is within **2.5 degrees** of the goal.

WHAT THE ROBOT CHECKS BEFORE EVERY SHOT, IN ORDER

CHECK	IF NOT, THE ROBOT SAYS	WHY
Is there a possible shot?	NO TRAJECTORY	do not waste pollen
Is the shooter calibrated?	CALIBRATION REQUIRED	simulation is not the real robot
Is it aimed and not still turning?	AIMING	within 2.5 degrees
Flywheels and hood on target?	SPINNING UP	within 50 RPM and 2 degrees
Ready for 0.1 s?	SETTLING	must stay on target, not pass through it
Is there pollen to shoot?	EMPTY	the beam sensors know
All yes	FEEDING	pollen goes to the shooter

MATHS: TURNING TO AIM

Turn speed = **feedback** on the heading error + **damping** + **feedforward** for a goal that moves as we drive + a small push past **static friction**.

Gains in our code: kP 4.0, kD 0.35, kS 0.06.

In simulation aiming first stopped **12 degrees** short: small commands were inside the static friction dead zone.

MATHS: SHOOTING WHILE MOVING

The pollen leaves with the shooter's speed **plus the robot's**:

v pollen = v shot + v robot + turn rate × r

(r = where the pollen leaves, from the robot's centre). The code looks **0.07 s** ahead to where the robot will be at release, then a gravity solver finds the speed and hood angle.

TESTS

TEST	SIMULATION	ROBOT
Score while still	3 of 3	not built yet
Score while moving	3 of 3	not built yet

OUR ROUTE

<auto route drawing: the team's hand-drawn route, redrawn cleanly>

EACH STEP

STEP	WHAT	POINTS
1	<step>	<pts>
2	<step>	<pts>
3	<step>	<pts>

STATUS

- The autonomous for this robot is **not written yet**: NOT TESTED
- <date we plan to have it running>

WHAT IT CAN USE

PART	WHAT IT GIVES THE AUTONOMOUS
Wheels + IMU	where we are as we drive
Rear camera	AprilTags, to know where we are on the field
Front camera	a path to the pollen
Beam sensors	how much pollen we hold
Auto-aim	the same shot checks as TeleOp (page 13)

TEST RESULTS

TEST	RESULT	STATUS
24 automated code tests	all pass	**PASS**
Simulation: score 3 of 3, still and moving	3 of 3	**PASS**
Simulation: pick up 3, aim, score 3	3 of 3	**PASS**
Simulation: lost heading sensor	stops drive and feeding	**PASS**
Robot: shots in from 3 distances	not built yet	**NOT TESTED**
Robot: time between shots	not built yet	**NOT TESTED**
Robot: top speed	not built yet	**NOT TESTED**
Robot: X-lock push	not built yet	**NOT TESTED**
Robot: jams per 20 pickups	not built yet	**NOT TESTED**
Autonomous	not written yet	**NOT TESTED**

LESSONS WE APPLIED

- A turret cost months last season, so we let the drivetrain aim
- Sketch the pollen path before CAD: sizes and spacing first
- One unit for each value in code (RPM, radians)
- Merge subsystems early: the bugs on page 12 only showed when they met

NEXT

- Build the robot, then run every test on this page
- Calibrate the shooter on the real robot
- Measure the lookup table
- Write and test the autonomous
- <next design change>

RELIABILITY

No matches yet: the robot is not built. For every match we will record whether it worked all match, and any fault and its fix.

Endnote: text drafted and laid out with help from an AI assistant. All results, photos, CAD, code and decisions are the team's own; the team checked every page.